



CINEMATCH

An AI-Powered Movie & Game Recommender

Describe your mood — get the perfect pick, and the reason why.

Software Engineering Capstone · Sejong University

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Too much choice, too little time

“

*People often spend **longer choosing** what to watch than they spend enjoying a short film.*

Decision fatigue is real.



Endless scrolling

Streaming menus offer hundreds of rows and no clear answer.



Optimized for engagement

Platforms maximise watch-time, not a confident decision.



No reason given

Recommendations rarely explain why a title suits you.

One answer. And the reason why.

CINEMATCH is a web app where you describe your mood in plain words — and an **AI** plus a **C engine** instantly pick the perfect movie or game for tonight, and tell you why.

Opinionated

One confident pick — not 20 rows of thumbnails.

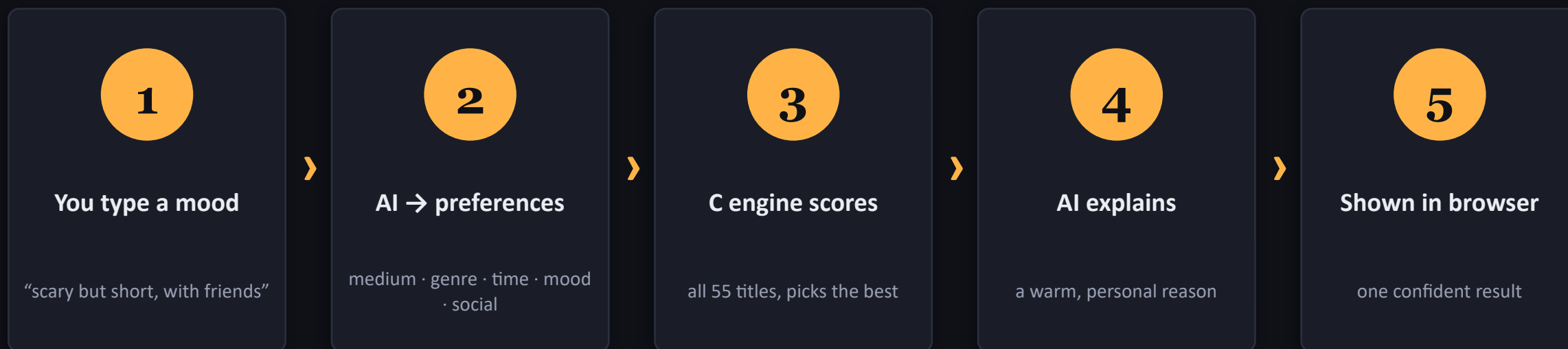
Explained

A warm, plain-language reason you can trust.

Hybrid

AI for language, C for deterministic logic.

The five-step pipeline



AI handles language · C handles deterministic logic

Domain-Driven Design

Separate the business idea from the plumbing — each layer has one job.

Interfaces

What users reach: web pages + HTTP API

Vite + React · Express

Application

Use cases: understand → pick → explain → discuss

Node.js

Domain

Core concepts & rules · the scoring

Node.js + C engine

Infrastructure

The outside world: AI client, engine runner

OpenRouter · child process

The Node backend runs the compiled C program as a child process, exchanging JSON.

An LLM at three points



Mood understanding

Reads free text and emits structured preferences as strict JSON — turning vague human language into machine-usable data.



Explanation generation

After the C engine picks a title, the LLM writes a warm two-to-three sentence reason, tailored to what the user said.



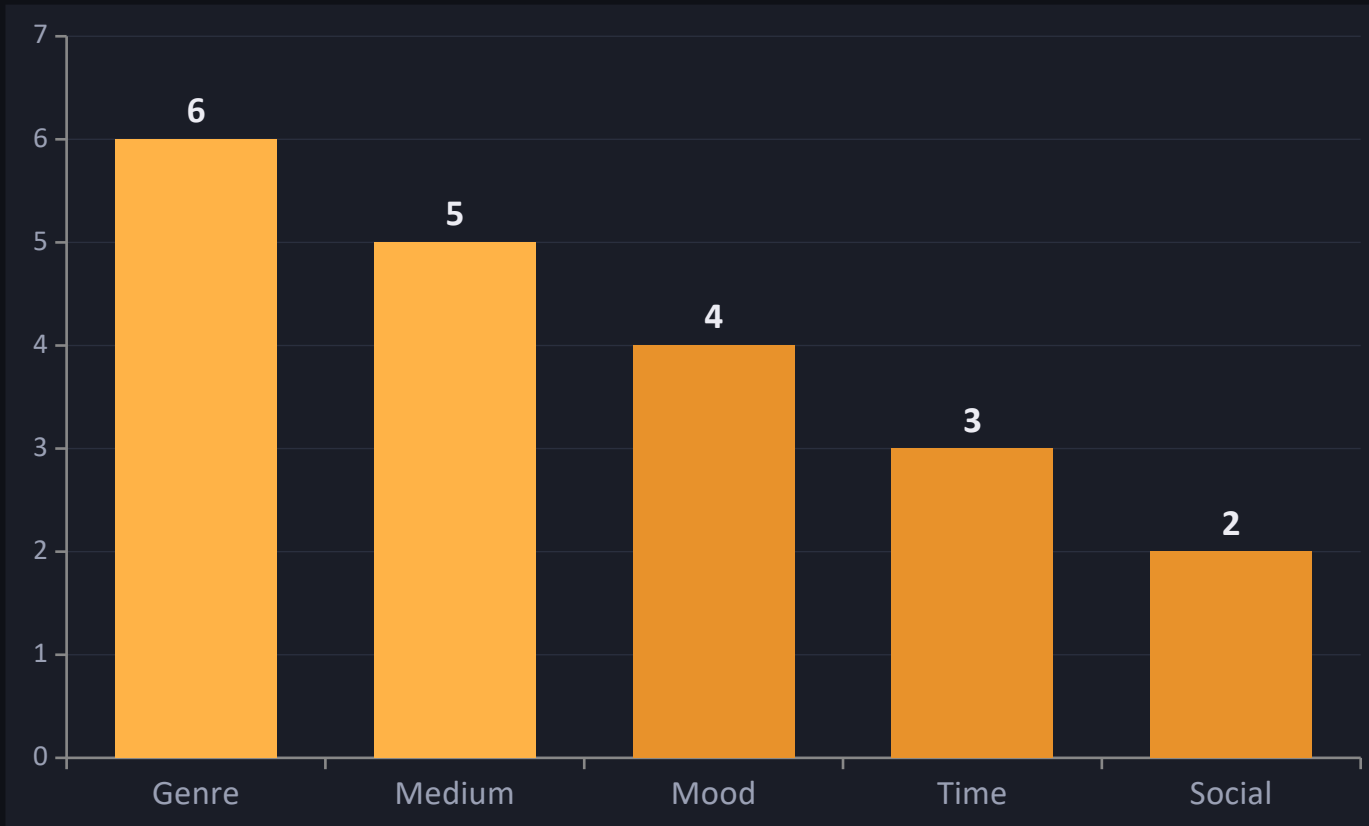
Movie discussion

Generates a full analysis of any film: theme, epic moments, director's vision, and discussion questions.

Provider: **OpenRouter — free open models (Qwen3)** · zero cost for students

Transparent, rule-based scoring

Each title earns points; the highest score wins. The biggest weights reflect what matters most.



Engine Mode

One C program, two modes:

No args

→ interactive 5-question quiz

5 args

→ prints one line of JSON

```
cinematch Movie horror short  
intense group
```

Four pages, one app



AI Match

The core feature. Type a mood, get one explained pick from the curated 55-title library.



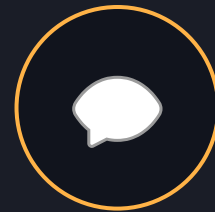
Movies

Browse thousands of real films from the open TMDB dataset — search, filter, posters, details.



Games

Browse 800,000+ titles from the open RAWG dataset — covers, ratings, and detail views.

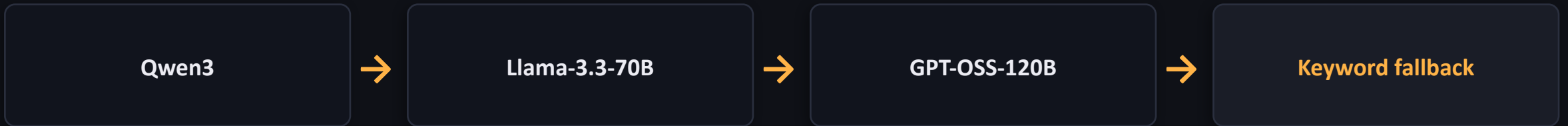


Discussion

Search any film for AI analysis: cast, theme, epic moments, and discussion questions.

Never crash in a live demo

Free AI models rate-limit and fail. CINEMATCH degrades gracefully at every layer.



Automatic retry

On a 429 or 5xx error, the request retries after a short delay.



Keyword fallback

If all AI fails, a local matcher still derives sensible preferences.



Content fallback

Empty AI fields are replaced with rich, well-written defaults.

Rehearsed and reliable

“relaxing cozy solo night”



Balatro (2024)

Cozy · Relaxed · ★ 9.0

“scary horror with a group”



Alien (1979)

Horror · Intense · ★ 8.5

“epic action game for hours”



Super Mario Odyssey

Action · Light · ★ 9.4

Same mood → same pick, every time. The C engine is deterministic.

Tech stack & open data

ENGINE

C → cinematch.exe

BACKEND

Node.js + Express

FRONTEND

Vite + React

PACKAGES

pnpm

AI

OpenRouter (Qwen3)

MOVIES

TMDB open dataset

GAMES

RAWG open dataset

VERSION CONTROL

Git + GitHub



Classic C meets modern AI.

A focused idea, executed cleanly — deterministic logic in C, language in an LLM,
wrapped in a polished, well-architected product.

Thank you

github.com/zubaydullayevasliddin06-cmd · CINEMATCH